

Defining our text examples

```
In[:text_to_search = '''
    abcdefghijklmnopqurtuvwxyz
    ABCDEFGHIJKLMNOPQRSTUVWXYZ
    1234567890
    Ha HaHa
    MetaCharacters (Need to be escaped):
    . ^ $ * + ? { } [ ] \ | ( )
    coreyms.com
    321-555-4321
    123.555.1234
    123*555*1234
    800-555-1234
    900-555-1234
    Mr. Schafer
    Mr Smith
    Ms Davis
    Mrs. Robinson
    Mr. T
    '''

    sentence = 'Start a sentence and end'

    emails = '''
    CoreyMSchafer@gmail.com
    corey.schafer@university.edu
    corey-321-schafer@my-work.net
    '''

    urls = '''
    https://www.google.com
    http://coreyms.com
    https://youtube.com
    https://www.nasa.gov
    '''

    data = '''
    Dave Martin
    615-555-7164
    173 Main St., Springfield RI 55924
    davemartin@bogusemail.com

    Charles Harris
    800-555-5669
    969 High St., Atlantis VA 34075
    charlesharris@bogusemail.com

    Eric Williams
    560-555-5153
    806 1st St., Faketown AK 86847
    laurawilliams@bogusemail.com
```

Getting started with Regular Expressions in Python

The re module

```
In [ ]:import re
In [ ]:print('\tTab')
        print(r'\tTab')

        Tab
\tTab
```

Diving into Regular Expressions

```
In [ ]:pattern = re.compile(r'abc')
        matches = pattern.search(text_to_search)
        print(matches)

        print(text_to_search[1:4])
<re.Match object; span=(1, 4), match='abc'>
abc
In [ ]:pattern = re.compile(r'ABC')
        matches = pattern.search(text_to_search)
        print(matches)
<re.Match object; span=(28, 31), match='ABC'>
```

Functions in the re module

Match vs Search vs Find

```
In [ ]:matches = re.match("c", "abcdef")
        print(matches)

None
In [ ]:matches = re.search("c", "abcdef")
        print(matches)

<re.Match object; span=(2, 3), match='c'>
In [ ]:matches = re.findall("c", "abcdef")
        print(matches)

['c']
In [ ]:matches = re.finditer("c", "abcdef")
        print(matches)

        for match in matches:
            print(match)

<callable_iterator object at 0x00000261BAB9ED40>
<re.Match object; span=(2, 3), match='c'>
```

Restricting matches at the beginning of a string

None

None

<re.Match object; span=(0, 1), match='a'>

Looping while pattern matching

```
In[:import time
```

```
In[:start = time.time()
```

```
    for i in range(1000):
        re.match('hello', 'hello world')
```

```
    end = time.time()
    print(end - start)
```

0.0020003318786621094

```
In[:start = time.time()
```

```
    h=re.compile('hello')
```

```
    for i in range(1000):
        h.match('hello world')
```

```
    end = time.time()
    print(end - start)
```

0.0

Meta Characters

```
In[:pattern1 = re.compile(r'\d')
```

```
    matches1 = pattern1.search(text_to_search)
```

```
    print (matches1)
```

<re.Match object; span=(55, 56), match='1'>

```
In[:pattern2 = re.compile(r'\bHa')
```

```
    matches = pattern2.finditer(text_to_search)
```

```
    for match in matches:
```

```
        print(match)
```

<re.Match object; span=(66, 68), match='Ha'>

<re.Match object; span=(69, 71), match='Ha'>

```
In[:pattern2 = re.compile(r'\BHa')
```

```
    matches = pattern2.finditer(text_to_search)
```

```
    for match in matches:
```

```
        print(match)
```

<re.Match object; span=(71, 73), match='Ha'>

```
In[:pattern = re.compile(r'^Start')
```

```
    matches = pattern.search(sentence)
```

```
    print(matches)
```

<re.Match object; span=(0, 5), match='Start'>

```
In[:pattern = re.compile(r'sentence')
```

```
    matches = pattern.search(sentence)
```

```
    print(matches)
```

<re.Match object; span=(8, 16), match='sentence'>

```
In[:
```

None

Practical examples

```
In [ ]: pattern = re.compile(r'\d\d\d.\d\d\d.\d\d\d\d')
        matches = pattern.search(text_to_search)
        print(matches)

<re.Match object; span=(151, 163), match='321-555-4321'>
In [ ]: pattern = re.compile(r'(M(r|s|rs)\.?\s[A-Z]\w*)')
        matches = pattern.findall(text_to_search)
        print(matches)
        # for match in matches:
        #     print(match)

[('Mr. Schafer', 'r'), ('Mr Smith', 'r'), ('Ms Davis', 's'), ('Mrs. Robinson', 'rs'), ('Mr. T', 'r')]
In [ ]: pattern = re.compile(r'\d\d\d[-.]\d\d\d[-.]\d\d\d\d')
        matches = pattern.finditer(data)
        for match in matches:
            print(match)

<re.Match object; span=(13, 25), match='615-555-7164'>
<re.Match object; span=(103, 115), match='800-555-5669'>
<re.Match object; span=(192, 204), match='560-555-5153'>
<re.Match object; span=(282, 294), match='900-555-9340'>
<re.Match object; span=(379, 391), match='714-555-7405'>
In [ ]: pattern = re.compile(r'[89]00[-.]\d\d\d[-.]\d\d\d\d')
        matches = pattern.finditer(data)
        for match in matches:
            print(match)

<re.Match object; span=(103, 115), match='800-555-5669'>
<re.Match object; span=(282, 294), match='900-555-9340'>
In [ ]: pattern = re.compile(r'[1-5]\d\d[-.]\d\d\d[-.]\d\d\d\d')

        matches = pattern.finditer(data)
        for match in matches:
            print(match)

<re.Match object; span=(192, 204), match='560-555-5153'>
In [ ]: pattern = re.compile(r'^b[at')
        matches = pattern.finditer("cat sat bat rat")
        for match in matches:
            print(match)

<re.Match object; span=(0, 3), match='cat'>
<re.Match object; span=(4, 7), match='sat'>
<re.Match object; span=(12, 15), match='rat'>
In [ ]: pattern = re.compile(r'\d{3}.\d{3}.\d{4}')
        matches = pattern.finditer(data)
        for match in matches:
            print(match)
```

```

<re.Match object; span=(13, 25), match='615-555-7164'>
<re.Match object; span=(103, 115), match='800-555-5669'>
<re.Match object; span=(192, 204), match='560-555-5153'>
<re.Match object; span=(282, 294), match='900-555-9340'>
<re.Match object; span=(379, 391), match='714-555-7405'>
In [ ]:pattern = re.compile(r'Mr\.\?\s[A-Z]\w*')
        matches = pattern.finditer(text_to_search)
        for match in matches:
            print(match)

<re.Match object; span=(216, 227), match='Mr. Schafer'>
<re.Match object; span=(228, 236), match='Mr Smith'>
<re.Match object; span=(260, 265), match='Mr. T'>
In [ ]:pattern = re.compile(r'M(r|s|rs)\.\?\s[A-Z]\w*')
        matches = pattern.finditer(text_to_search)
        for match in matches:
            print(match)

<re.Match object; span=(216, 227), match='Mr. Schafer'>
<re.Match object; span=(228, 236), match='Mr Smith'>
<re.Match object; span=(237, 245), match='Ms Davis'>
<re.Match object; span=(246, 259), match='Mrs. Robinson'>
<re.Match object; span=(260, 265), match='Mr. T'>

```

Other

```

In [ ]:pattern = re.compile(r'https?:/(www\.)?(\w+)(\.\w+)')

        subbed_url = pattern.search(urls)
        subbed_urls = pattern.findall(urls)

        print(subbed_url)

        for match in subbed_urls:
            print(match)

<re.Match object; span=(1, 23), match='https://www.google.com'>
('www.', 'google', '.com')
('', 'coreyms', '.com')
('', 'youtube', '.com')
('www.', 'nasa', '.gov')
In [ ]:pattern = re.compile(r'M(rs|r|s)\.\? [A-Z][a-z]*')
        matches = pattern.finditer(text_to_search)
        for match in matches:
            print(match)

<re.Match object; span=(216, 227), match='Mr. Schafer'>
<re.Match object; span=(228, 236), match='Mr Smith'>
<re.Match object; span=(237, 245), match='Ms Davis'>
<re.Match object; span=(246, 259), match='Mrs. Robinson'>
<re.Match object; span=(260, 265), match='Mr. T'>
In ...pattern = re.compile(r'[A-Za-z]+([\.-]?[A-Za-z0-9]+)+@[A-Za-z]+[\-]?[A-Za-z0-9]+?\.'
        matches = pattern.finditer(emails)
        for match in matches:
            print(match)

```

```
<re.Match object; span=(1, 24), match='CoreyMSchafer@gmail.com'>  
<re.Match object; span=(25, 53), match='corey.schafer@university.edu'>  
<re.Match object; span=(54, 83), match='corey-321-schafer@my-work.net'>  
In [ ]: pattern = re.compile(r'a(a|b|c)+')  
        matches = pattern.findall(string)  
        for match in matches:  
            print(match)
```

a